CAMERAS

A LIGHTPROOF BOX with a hole or lens at one end, and a strip of light-sensitive film at the other, is the basic component of a traditional camera. To take a photograph, the photographer points the camera at an object and presses a button. This button very briefly opens a shutter behind the lens. Light reflected from the object passes through the lens and on to traditional film or a digital chip to produce an image.

Parts of a camera

The quality of a photograph is controlled by adjusting the film and shutter speed dials, flash, and aperture scales. This is because the final image will depend on the type of film in the camera, the amount of light that enters the lens, and the length of time that the film is exposed to light.

35mm cameras

The most popular cameras are the 35mm, named after the width of the film they use. These cameras are small and easy to manage. They often have in-built features, which adjust automatically to variations in light and distance, to ensure that a clear photograph is taken every time.

Computer imaging

After an image has been stored on a digital camera, it can then be fed into a computer. From here it is printed out on photo paper or sent over the Internet. Special software allows the picture to be manipulated and gives the photographer a lot of control over the image.

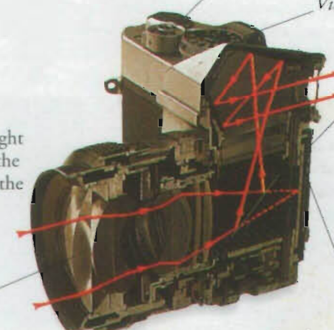
Digital cameras

Digital cameras contain no film. Instead, the image is captured on a photosensitive chip. Photos are displayed instantly on a screen on the camera and can be deleted if not liked. Images can be loaded into a computer and printed out.

Digital camera

Flashes

A flash provides the extra light needed for taking pictures after dark, or in dim conditions. The flash is electronically controlled to go off at the moment the shutter opens.



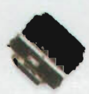
As the shutter is released, the mirror slips up allowing the light to reach the film (shown by the dotted line).

Single-lens reflex camera

Unlike other cameras, the view through a single-lens reflex (SLR) camera is that of the actual image that is recorded on the film. Mirrors in the viewfinder correct the upside-down image sent from the lens.

Lenses

Different lenses achieve different visual effects. A wide-angle lens allows more of the scene to appear in a photograph than a normal lens. A telephoto zoom lens can take a close-up shot of a distant object. The fisheye lens distorts images for dramatic effect. These lenses are detachable from the camera.



Normal lens

Telephoto zoom lens

Wide-angle lens

Fisheye lens

Film types

Today, plastic film comes in various sizes and speeds, in a colour or a black and white format, packaged as rolls or plates. The speed, given in ASA/ISO or DIN numbers, indicates how quickly the film reacts to light. A new device, the Electronic Film System, fits into a 35mm camera and holds up to 30 digital images which can be transferred to a computer.

110mm film

35mm film

Plate film

George Eastman

An American inventor, George Eastman (1854–1932), formed the Kodak company. In 1884, he produced the first roll film and in 1888 the first box camera, making photography an accessible hobby. In 1889, he used clear celluloid film on which the first movie pictures were taken.

Timeline

4th century BC The "camera obscura" is developed; it consists of a darkened room into which an image is projected.

1822 Frenchman Joseph Niepce takes the first photograph on a sheet of pewter, coated with bitumen.

1839 Niepce's colleague, Louis Daguerre, announces process for recording images on copper.

1839 William Fox Talbot, an Englishman, invents a process that allows photographs to be copied.

1895 The Lumière brothers of France patent their original camera/projector using celluloid film with sprocket holes at the edge.

1948 American inventor Edwin Land developes the first instant camera, which is marketed by the Polaroid Corporation.

1956 A camera that records onto reel-to-reel magnetic videotape, rather than plastic film, is invented.

1980s First digital cameras prototyped.

1986 Disposable camera launched.

1992 The jpeg, a compressed file format for storing digital images, is introduced.

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